CALIBO AI ACADEMY · PATH 2 · KLU BATCH · MUC02

Credit Pre-Screening Engine

Loan Intake Optimisation — Retail Personal Lending
Supervised Learning Part 1 · Corrected Team Analysis Report

Team and Role Allocation

Team member	Sections	Responsibilities
G. Akash ID 2300080420	CBIM, Steps 5.1–5.2	CBIM coordination; dataset loading, inspection and cleaning; historical approval analysis by credit-score band and employment type.
D. Siva Danesh ID 2300040119	Step 5.3	Feature engineering; FOIR and loan-to-income calculations; NTC identification and retention.
R. Devisree ID 2300080412	Steps 5.4–5.5	Logistic Regression, model evaluation, coefficients, threshold selection and monthly business-impact analysis.
B. Varshini ID 2300080058	Steps 5.6–5.7	Fairness audit by gender and employment type; proxy-risk analysis; credit-policy briefing and recommendations.

Executive Summary

12,000 APPLICATIONS	**47.1%** HISTORICALLY DECLINED	**78.0%** TEST ACCURACY · 0.50	**0.95** SELECTED THRESHOLD
9.62% TEST AUTO-DECLINE	**1.30%** TEST WRONG-DECLINE	**~289** APPLICATIONS / MONTH	**~120.3** HOURS / MONTH

The team analysed 12,000 historical personal-loan applications (April–September 2026) to build a Logistic Regression pre-screener. The business objective is to identify a conservative auto-decline lane while keeping the share of historically approved applicants in that lane within 5%.

Corrected result: The validation set selected threshold **0.95**. On the separate held-out test set, 9.62% of applications entered the auto-decline lane and 1.30% of those had historically been approved. At 3,000 monthly applications and 25 minutes per review, this represents approximately 289 applications and 120.3 analyst hours per month.

CBIM Problem Canvas

Element	Description
Situation	About 3,000 personal-loan applications arrive monthly; each receives about 25 minutes of manual review. Historical data shows 47.1% were declined.
Complication	Analyst capacity is spent reviewing applications with clear historical risk signals, while other applicants wait.
Question	Which applications can safely leave manual review through a conservative auto-decline lane, and at what threshold?
Single source of truth	MUC02_Credit_PreScreening_Dataset.csv: 12,000 applications and 19 original columns. Decision records past analyst judgement, not repayment.
Success definition	Select a threshold on validation data with no more than 5% historically approved applicants among auto-declines; evaluate once on held-out test data.
Stakeholder	Credit Policy Analyst; credit analysts and applicants are also affected.

Step 5.1 — Load, Inspect and Clean

The dataset contains 12,000 rows and 19 original columns. There are no duplicate rows or application IDs. The target distribution is 6,353 Approved (52.9%) and 5,647 Declined (47.1%).

Issue	Count	Treatment / reason
Missing monthly income	441	Imputed using employment-type median.
Missing credit score	1,039 (8.7%)	Retained and flagged as NTC; all have credit_history_months = 0.
Unusually high salaried income	12	Divided by 12; likely annual income entered as monthly.

Data-quality outcome: all 12,000 applications retained; 441 income values imputed; 1,039 NTC applicants preserved; no duplicate rows or IDs.

Methodological limitation: The notebook calculates employment-type income medians before splitting. This allows validation/test information to influence imputation. Recalculate medians on training data only and rerun downstream outputs for a leakage-safe final model.

Steps 5.2–5.3 — Historical Approval and Affordability

Credit-score approval increases sharply at 650–699. Applicants without a score have a higher approval rate than low-score applicants and should not be rejected solely for missing credit history.

Credit-score band	Approval rate
Below 600	2.0%
600–649	3.2%
650–699	32.5%
700–749	54.4%
750–799	77.2%
800+	85.2%
No score	53.9%

Employment type	Approval rate
Salaried	58.4%
Business Owner	51.2%
Self-Employed Professional	50.6%
Gig / Contract	34.9%

Affordability: FOIR = (existing EMI + proposed EMI) ÷ monthly income. Loan-to-income = loan amount ÷ (monthly income × 12). The NTC flag equals 1 when credit score is missing.

FOIR band	Approval rate
Below 30%	65.5%
30–40%	59.0%
40–50%	47.8%
50–55%	32.4%
Above 55%	15.6%

Approval declines as FOIR rises, with a pronounced fall above 50%. A 50–55% band may prompt additional affordability review; it is not a proven safe lending limit because the target is historical analyst judgement, not repayment.

Step 5.4 — Logistic Regression Pre-Screener

Target: Declined = 1; Approved = 0. Twelve features were used: credit_score, is_ntc, foir, loan_to_income, delinquencies_24m, num_active_loans, years_in_current_job, age, tenure_months, city_tier, employment_type and loan_purpose. ID, date, city, gender and decision were excluded.

The notebook uses a stratified 60/20/20 split: 7,200 training, 2,400 validation and 2,400 test applications. Numeric features are imputed and scaled; employment type and loan purpose are one-hot encoded in the pipeline.

78.0%	0.867	78.2%	73.7%
TEST ACCURACY	ROC-AUC	PRECISION · DECLINED	RECALL · DECLINED

Confusion Matrix — Test Set at 0.50

Actual class	Predicted Approved	Predicted Declined
Actually Approved	1,039 correctly approved	232 wrongly declined
Actually Declined	297 declines missed	832 correctly declined

At threshold 0.50, 232 applicants historically approved by the credit team would be classified as Declined. Accuracy alone is therefore not an adequate business measure.

Largest Logistic Regression Coefficients

Feature	Coefficient	Interpretation
FOIR	+1.269	Higher EMI burden is associated with greater decline probability.
Credit score	–1.003	Higher score is associated with lower decline probability.
Delinquencies (24m)	+0.982	More delinquencies are associated with greater decline probability.
Gig / Contract	+0.655	Associated with greater decline probability.
Salaried	–0.425	Associated with lower decline probability.
NTC flag	+0.390	Associated with greater decline probability in this fitted model.

Coefficients describe model associations, not causal effects. NTC applicants should not be auto-declined: model accuracy for this group is only 56.7%.

Step 5.5 — Threshold Selection and Business Case

Threshold selection is performed on validation data only. NTC applicants are excluded from automatic decline. The lowest tested threshold meeting the maximum 5% wrong-decline share is 0.95.

Threshold	Auto-declined %	Wrong-decline share	5% gate
0.50	41.00%	19.31%	Fail
0.60	34.71%	14.89%	Fail
0.70	28.71%	10.45%	Fail
0.80	22.79%	7.50%	Fail
0.85	18.92%	6.39%	Fail
0.90	14.92%	5.03%	Fail
0.95	9.88%	2.53%	Pass

Threshold 0.90 fails the validation gate (5.03% exceeds 5%). The validation-selected threshold is 0.95.

Held-Out Test Results at 0.95

Measure	Test result
Auto-declined	231 of 2,400 (9.62%)
Historically approved within auto-declines	3 of 231
Wrong-decline share	1.30%
Share of all historical declines caught	20.19%

Monthly Business Estimate

Measure	Estimate
Monthly applications	3,000
Review time per application	25 minutes
Applications avoiding manual review	~289 per month
Analyst hours saved	~120.3 per month

Calculation: 3,000 × 9.62% ≈ 289 applications; 289 × 25 ÷ 60 ≈ 120.3 hours. These are estimates, not guaranteed savings. The test-set wrong-decline share is a sample estimate and may change on future applications.

Step 5.6 — Fairness Audit

Gender is excluded from model features and used only for post-model monitoring. These results are from the held-out test set of 2,400 applications, using threshold 0.95 and the NTC human-review safeguard.

Test-set indicator	Female	Male
Applications	714	1,686
Historical approval rate	51.12%	53.74%
Median monthly income	■46,000	■56,500
Gig / Contract share	20.45%	11.51%
Model auto-decline rate	10.64%	9.19%

Historical Approval Within Employment Type — Test Set

Employment type	Female	Male
Business Owner	48.33%	51.52%
Gig / Contract	35.62%	34.02%
Salaried	56.97%	58.72%
Self-Employed Professional	51.52%	51.27%

The test set shows a 2.62 percentage-point difference in historical approval rates and a 1.45 percentage-point difference in model auto-decline rates. Female applicants in this test sample have lower median income and a higher Gig / Contract share. These are observed associations; the analysis does not establish causation.

All employment-by-gender subgroups have at least 30 test cases. This minimum does not prove fairness; continue monitoring approval, auto-decline, false-decline and repayment outcomes.

Excluding gender is necessary but not sufficient. Income and employment may act as proxies. Keep NTC applicants in manual review and monitor subgroup outcomes.

Step 5.7 — Credit Policy Briefing

"I recommend piloting an auto-decline threshold of 0.95, selected on validation data because it is the lowest tested threshold meeting the 5% wrong-decline limit. On the held-out test set, this threshold auto-declined 9.62% of applications, with 1.30% of those historically approved by the credit team. At 3,000 applications per month and 25 minutes per review, this represents approximately 289 applications and 120 analyst hours of potential monthly capacity. FOIR (+1.269) and credit score (–1.003) are the strongest model coefficients, while the model should not independently decide on new-to-credit applicants because its accuracy for that group is only 56.7%. Gender is excluded, but test-set auto-decline rates differ (10.64% female and 9.19% male), so subgroup outcomes and proxy effects require monitoring. Before production use, I would require repayment/default outcomes, out-of-time validation, calibration checks and an approved human review and appeal process."

Recommendations, Risks and Limitations

Category	Action / finding
Recommendation	Pilot 0.95 on controlled volume; do not treat it as a final lending decision.
NTC safeguard	Keep NTC applicants in manual review; subgroup accuracy is 56.7%.
Affordability	Use FOIR above 50% as a prompt for additional assessment, not automatic rejection.
Fairness	Monitor gender and employment subgroup outcomes and investigate proxy effects.
Risk	The 1.30% wrong-decline share is a test-sample estimate, not a guarantee.
Target limitation	The model predicts past analyst decisions, not actual repayment.
Data limitation	Income medians are currently calculated before splitting; use train-only imputation and rerun.
Further validation	Obtain repayment/default data; conduct out-of-time validation; assess calibration and drift; document human review and appeal.

Conclusion: The notebook supports a cautious 0.95 screening pilot with an estimated 120 analyst hours saved per month. It does not support unsupervised lending automation.

This corrected edition aligns the report's model metrics, confusion matrix, coefficients, threshold, business case and fairness figures with the saved outputs in the supplied notebook.